

## Problem-Solution Fit

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|----------------------|------------------------------------------------------------|
| <b>Date</b>          | 15 March 2026                                              |
| <b>Team ID</b>       | XXXXXX                                                     |
| <b>Project Name</b>  | Opticrop-Smart Agricultural Production Optimization Engine |
| <b>Maximum Marks</b> | 5 Marks                                                    |

### Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

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| <b>1. CUSTOMER SEGMENT(S) [CS]</b> <ul style="list-style-type: none"> <li>• Farmers</li> <li>• Agricultural Researchers</li> <li>• Agricultural Policymakers</li> <li>• Agricultural Extension Officers</li> </ul>                                                                                                                                                                                                                                                                                                           | <b>6. CUSTOMER LIMITATIONS</b><br><b>EG. BUDGET, DEVICES [CL]</b> <ul style="list-style-type: none"> <li>• Limited technical knowledge</li> <li>• Limited access to advanced tools</li> <li>• Budget constraints</li> <li>• Dependence on local environmental conditions</li> </ul>                                                                                                                                                                                                                                                                                                                      | <b>5. AVAILABLE SOLUTIONS</b><br><b>PROS &amp; CONS [AS]</b> <ul style="list-style-type: none"> <li>• <b>Traditional Farming Methods</b> <ul style="list-style-type: none"> <li>– Pros: Easy to follow, experience-based</li> <li>– Cons: Less accurate, not data-driven</li> </ul> </li> <li>• <b>Agricultural Experts</b> <ul style="list-style-type: none"> <li>– Pros: Professional guidance</li> <li>– Cons: Limited availability and cost</li> </ul> </li> <li>• <b>Online Resources</b> <ul style="list-style-type: none"> <li>– Pros: Easily accessible</li> <li>– Cons: Generic recommendations</li> </ul> </li> </ul> |              |                                                                                                                                                        |                                                                                                                                                                                                 |
| <b>2. PROBLEMS / PAINS</b><br><b>+ ITS FREQUENCY [PR]</b> <ul style="list-style-type: none"> <li>• Difficulty selecting suitable crops (High)</li> <li>• Low crop yield due to poor decision-making (High)</li> <li>• Inefficient use of fertilizers and water (Medium)</li> <li>• Lack of data-driven farming insights (High)</li> <li>• Unpredictable impact of environmental conditions (High)</li> </ul>                                                                                                                 | <b>9. PROBLEM ROOT / CAUSE [RC]</b> <ul style="list-style-type: none"> <li>• Lack of scientific crop selection methods</li> <li>• Inability to analyze multiple environmental factors simultaneously</li> <li>• Dependence on traditional farming practices</li> <li>• Limited access to predictive agricultural tools</li> </ul>                                                                                                                                                                                                                                                                        | <b>7. BEHAVIOR</b><br><b>+ ITS INTENSITY [BE]</b> <ul style="list-style-type: none"> <li>• Regularly checks weather and soil conditions (High)</li> <li>• Uses traditional farming practices (High)</li> <li>• Seeks advice from experts and peers (Medium)</li> <li>• Adopts new technology cautiously (Medium)</li> <li>• Tries to reduce cost and increase yield (High)</li> </ul>                                                                                                                                                                                                                                           |              |                                                                                                                                                        |                                                                                                                                                                                                 |
| <b>3. TRIGGERS TO ACT [TR]</b> <ul style="list-style-type: none"> <li>• Declining crop productivity</li> <li>• Increasing farming costs</li> <li>• Unpredictable weather conditions</li> <li>• Need for higher profits</li> <li>• Awareness about smart farming technologies</li> </ul>                                                                                                                                                                                                                                      | <b>10. YOUR SOLUTION [SL]</b><br><b>OptiCrop – Smart Agricultural Production Optimization Engine</b> <ul style="list-style-type: none"> <li>• Uses Machine Learning to analyze soil nutrients (N, P, K) and environmental factors such as temperature, humidity, pH, and rainfall.</li> <li>• Predicts the most suitable crop for higher yield and profitability.</li> <li>• Optimizes resource usage (water, fertilizers, land).</li> <li>• Provides data-driven insights and easy-to-use web interface for farmers.</li> <li>• Supports sustainable and intelligent agricultural practices.</li> </ul> | <b>8. CHANNELS OF BEHAVIOR [CH]</b><br><b>ONLINE</b> <ul style="list-style-type: none"> <li>• Agricultural websites</li> <li>• Mobile applications</li> <li>• YouTube farming channels</li> <li>• Social media / Farming communities</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                 |              |                                                                                                                                                        |                                                                                                                                                                                                 |
| <b>4. EMOTIONS</b><br><b>BEFORE / AFTER [EM]</b> <table border="0"> <tr> <td><b>BEFORE</b></td> <td><b>AFTER</b></td> </tr> <tr> <td> <ul style="list-style-type: none"> <li>• Confused</li> <li>• Uncertain</li> <li>• Worried about losses</li> <li>• Frustrated with low yield</li> </ul> </td> <td> <ul style="list-style-type: none"> <li>• Confident in crop selection</li> <li>• Informed and aware</li> <li>• Optimistic about higher yield</li> <li>• Satisfied with better profits</li> </ul> </td> </tr> </table> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>BEFORE</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>AFTER</b> | <ul style="list-style-type: none"> <li>• Confused</li> <li>• Uncertain</li> <li>• Worried about losses</li> <li>• Frustrated with low yield</li> </ul> | <ul style="list-style-type: none"> <li>• Confident in crop selection</li> <li>• Informed and aware</li> <li>• Optimistic about higher yield</li> <li>• Satisfied with better profits</li> </ul> |
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